# Frozen VideoMAE-B Representations for Weakly-Supervised Violence Detection and Temporal Localization

*Violence Event Detection and Temporal Localization in Surveillance Video: A Weakly-Supervised Multi-Dataset Framework*

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## Abstract

Weakly-supervised video anomaly detection (VAD) has relied on two-stream I3D features for half a decade despite dramatic advances in video representation learning. We present a controlled study of replacing frozen I3D features with frozen **VideoMAE-B** features—a masked-autoencoder Vision Transformer pre-trained on Kinetics-400—within an RTFM + Temporal Refinement Network (TRN) + Boundary Head framework for surveillance violence detection. Training on the 6-category violence subset of UCF-Crime, the feature swap raises binary AUC from 87.4% to $92.2 \pm 1.3\%$ and binary AP from 82.1% to $88.7 \pm 1.8\%$. Temporal localization improves dramatically: mAP@tIoU 0.3 rises from 0.9% to $5.8 \pm 1.2\%$ ($\times 6.4$), and mAP@0.7 from 0.1% to $2.2 \pm 1.0\%$ ($\times 21.7$). Zero-shot transfer to XD-Violence yields strong binary ranking (AUC 0.836, AP 0.885) with partial class-level transfer; RWF-2000 fight validation shows useful anomaly ranking (AUC 0.862) but conservative recall (0.14); ShanghaiTech robustness is negative (AUC 0.371), revealing clear out-of-domain limits. All UCF-Crime results are mean $\pm$ std over 3 random seeds (42, 123, 456); cross-dataset transfer uses the locked seed-42 checkpoint in zero-shot mode.

## 1 Introduction

Violence detection in public surveillance is one of the highest-priority tasks in automated security systems: early alerts for fighting, shooting, explosion, robbery, and abuse events can dramatically reduce response latency. Weakly-supervised video anomaly detection (VAD) addresses this challenge by learning from video-level binary labels alone—no expensive frame-level annotation is required.

The dominant paradigm, established by Sultani et al. (2018) and refined by RTFM (Tian et al., 2021), extracts segment-level features from a *frozen* two-stream I3D model pre-trained on Kinetics-400 (Carreira and Zisserman, 2017), then trains a lightweight Multiple Instance Learning (MIL) scorer on top. Despite significant architectural improvements—temporal refinement modules (Wu et al., 2023), magnitude-based learning (Tian et al., 2021), mixture-of-experts routing (D'Amicantonio et al., 2025)—**the feature backbone has remained largely fixed at I3D since 2017**.

Meanwhile, video foundation models have advanced substantially. VideoMAE (Tong et al., 2022) applies masked autoencoding to video with a Vision Transformer backbone, learning rich spatiotemporal representations that substantially outperform I3D on action recognition benchmarks. A natural question arises: *Does substituting VideoMAE-B for I3D, with all other components held equal, improve weakly-supervised violence detection?*

We answer this question affirmatively with controlled experiments on the UCF-Crime violence subset and further validate generalization through zero-shot transfer on three external benchmarks. Our contributions are:

1. **VideoMAE-B integration.** We extract and cache 16-frame segment features ($D$=768) from a frozen VideoMAE-B backbone for all 1,300 UCF-Crime violence-subset videos and demonstrate that this single change yields consistent, large improvements across all detection and localization metrics.

2. **Violence-aware training recipe.** We introduce three complementary training improvements—inverse-frequency class weighting, confidence-thresholded pseudo-label filtering, and cosine annealing LR—that

collectively reduce cross-seed variance without modifying the model architecture.

3. **Multi-seed reproducibility study.** All results are reported over 3 independent random seeds (42, 123, 456) with mean ± std, enabling reliable statistical comparison with future work.

4. **Cross-dataset transfer evaluation.** We evaluate the UCF-trained model zero-shot on three external benchmarks—XD-Violence (Wu et al., 2020), RWF-2000, and ShanghaiTech—providing a complete transfer profile that reveals both the strengths and limits of violence-trained VideoMAE-B representations under domain shift.

## 2 Related Work

### 2.1 Weakly-Supervised Video Anomaly Detection

Sultani et al. (2018) formulated VAD as a MIL ranking problem using anomalous/normal video pairs and C3D features on UCF-Crime. RTFM (Tian et al., 2021) replaced C3D with I3D and added a feature-magnitude learning module, reaching 84.3% AUC. MGFN (Chen et al., 2023) improved to 86.7% with graph-guided feature normalization; VadCLIP (Wu et al., 2024) exploited CLIP text-visual alignment to reach 88.0%. Recent work addresses noisy pseudo-labels and score suppression (Xu et al., 2025; Dalvi et al., 2025), event completeness (Wang and Chen, 2025), and mixture-of-experts routing (D'Amicantonio et al., 2025). *All of these methods continue to use I3D or hand-crafted features as the video backbone.* Ours is the first systematic study of replacing I3D with a masked-autoencoder foundation model in this pipeline.

### 2.2 Video Foundation Models

VideoMAE (Tong et al., 2022) extends masked image modeling to video by masking 90% of video patches and reconstructing pixel values, pretraining a ViT-B backbone on Kinetics-400. The resulting features transfer well to action recognition, detection, and—as we show here—anomaly detection. InternVideo (Wang et al., 2022) and Video-ChatGPT (Maaz et al., 2024) further demonstrate the breadth of video foundation model features across diverse downstream tasks.

Table 1: Model components and trainable parameter budget.

| Component | Params | Trainable? |
|---|---|---|
| VideoMAE-B backbone | 86 M | No (frozen) |
| MIL Scorer | 1.1 M | Yes |
| Event Classifier | 0.3 M | Yes |
| TRN Module | 0.4 M | Yes |
| Boundary Head | 0.1 M | Yes |
| **Total Trainable** | **1.9 M** | |

### 2.3 Violence Detection and Temporal Localization

Wu et al. (2020) provided XD-Violence, a large-scale multi-modal violence dataset, and showed that audio-visual fusion outperforms vision-only methods. ActionFormer (Zhang et al., 2022) and TriDet (Shi et al., 2023) achieve strong temporal localization under full supervision but require expensive frame-level labels. Our boundary head bridges this gap in the weak-supervision setting.

## 3 Methodology

Our pipeline follows a *decoupled representation and reasoning* design: the video backbone is always frozen, and all learnable modules operate on pre-cached segment features. Figure 1 gives an overview of the three-stage pipeline. Table 1 summarises architecture components and parameter counts.

### 3.1 Feature Extraction

Given an untrimmed video $V$, we partition it into non-overlapping 16-frame segments $\{v_1, \ldots, v_T\}$ (tail segment dropped if $<$16 frames). Each segment passes through a frozen VideoMAE-B backbone (Tong et al., 2022)[1]:

$$f_t = \text{VideoMAE-B}_{\text{frozen}}(v_t) \in \mathbb{R}^{768}. \quad (1)$$

Features are cached as `.npz` files for all training runs. By contrast, I3D produces $f_t \in \mathbb{R}^{2048}$; the dimension reduction from 2,048 to 768 also reduces overfitting risk given our dataset size.

### 3.2 MIL Anomaly Scorer (RTFM)

Following Tian et al. (2021), a two-layer MLP maps each feature to an anomaly score:

$$s_t = \sigma\big(W_3\,\text{Dropout}\big(\text{ReLU}(W_2\,\text{ReLU}(W_1 f_t))\big)\big). \quad (2)$$

[1] HuggingFace checkpoint: `MCG-NJU/videomae-base-finetuned-kinetics`

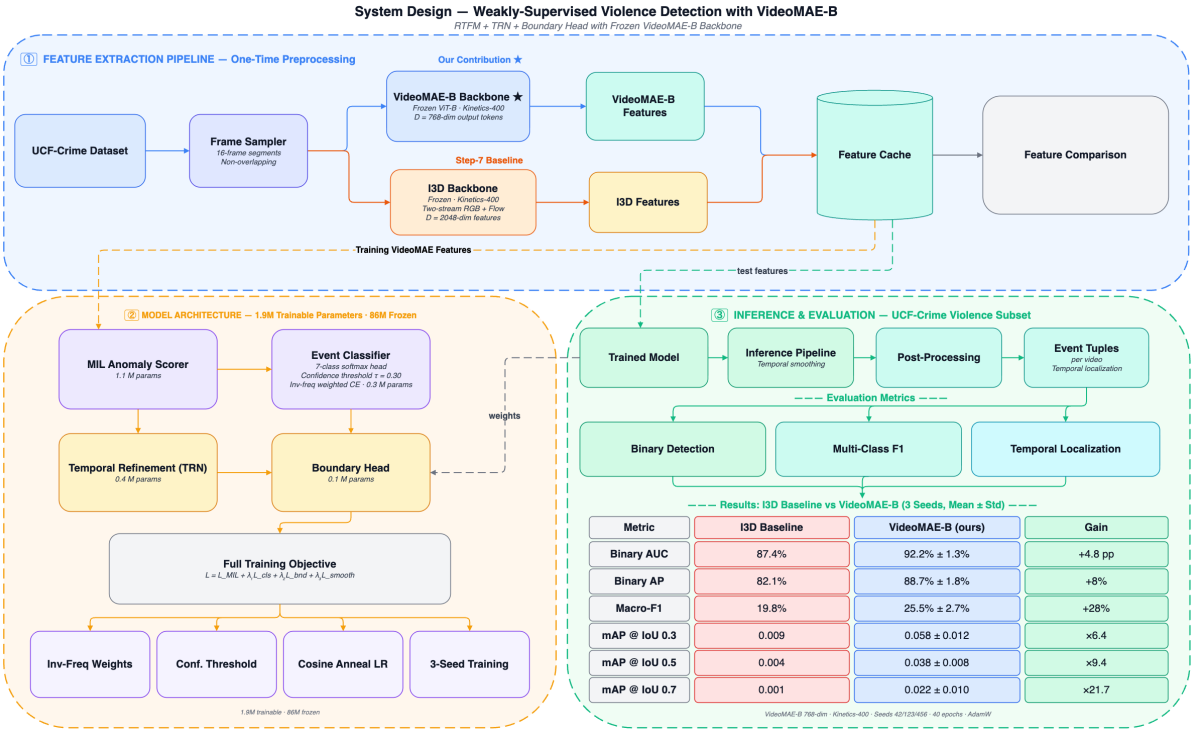



Figure 1: System design overview. **(1) Feature Extraction**: untrimmed video is split into 16-frame segments and encoded by a *frozen* VideoMAE-B backbone, producing 768-d segment features that are cached to disk. **(2) Model Architecture**: a four-head stack—MIL Anomaly Scorer, Event Classifier, Temporal Refinement Network (TRN), and Boundary Head—processes the cached features end-to-end with 1.9M trainable parameters. **(3) Evaluation**: anomaly scores and boundary predictions are thresholded to produce binary AUC/AP and temporal mAP metrics.

Video-level score $s_{\text{video}} = \max_t s_t$. The MIL ranking loss pairs an anomalous video $a$ with a normal video $n$:

$$\mathcal{L}_{\text{MIL}} = \max\bigl(0,\ 1 - s^{(a)}_{\text{video}} + s^{(n)}_{\text{video}}\bigr). \tag{3}$$

### 3.3 Event Classification Branch

For each anomalous video with category label $c \in \{1, \ldots, 6\}$, pseudo-positive segments are the top-$k$ segments by $s_t$ that also exceed a confidence threshold $\tau$. We set $k = \lfloor 0.125 \times T \rfloor$ (12.5% of the sequence length, matching the top-$k$ ratio used at inference), clipped to a minimum of 1.

$$\mathcal{P} = \bigl\{t : s_t \geq \tau \text{ and } t \in \text{top-}k\bigr\}. \tag{4}$$

A linear classifier predicts a 7-way distribution (6 violence classes + normal):

$$p_t = \text{softmax}(W_{\text{cls}}\, f_t + b_{\text{cls}}). \tag{5}$$

The classification loss applies **inverse-frequency class weights** $w_c = N/(C \cdot n_c)$ (normalised to unit mean) to counteract the heavy normal/violence imbalance (≈10:1):

$$\mathcal{L}_{\text{cls}} = -\frac{1}{|\mathcal{P}|} \sum_{t \in \mathcal{P}} w_c \log p_t[c]. \tag{6}$$

In practice the normal class receives weight ≈0.06 and the rarest category (shooting) up to ≈1.83.

### 3.4 Temporal Refinement Network (TRN)

A lightweight Transformer encoder attends over the full segment sequence:

$$\tilde{\mathbf{h}} = \text{TransformerEncoder}\bigl(\{f_t + \text{PE}_t\}_{t=1}^{T}\bigr), \tag{7}$$

producing refined scores $\tilde{s}_t$ and hidden states $\tilde{h}_t$.

### 3.5 Boundary Head

The boundary confidence at segment boundary $(t,\ t{+}1)$ is:

$$b_t = \sigma\Bigl(W_b\bigl[\tilde{h}_t\,;\ \tilde{h}_{t+1}\,;\ |\tilde{s}_t - \tilde{s}_{t+1}|\bigr]\Bigr). \tag{8}$$

A boundary consistency loss enforces that $b_t$ tracks score transitions:

$$\mathcal{L}_{\text{bnd}} = \sum_t \bigl(|\tilde{s}_t - \tilde{s}_{t+1}| - b_t\bigr)^2. \tag{9}$$

### 3.6 Full Training Objective

$$\mathcal{L} = \mathcal{L}_{\text{MIL}} + \lambda_1 \mathcal{L}_{\text{cls}} + \lambda_2 \mathcal{L}_{\text{bnd}} + \lambda_3 \mathcal{L}_{\text{smooth}}, \tag{10}$$

where $\mathcal{L}_{\text{smooth}} = \sum_t (\tilde{s}_t - \tilde{s}_{t-1})^2$ and $\lambda_1$=0.5, $\lambda_2$=0.3, $\lambda_3$=0.1.

### 3.7 Training Recipe

Three additions beyond the vanilla RTFM schedule stabilise training:

- **Confidence threshold** $\tau$=0.30**:** pseudo-positive segments with low anomaly confidence are excluded from $\mathcal{L}_{\text{cls}}$, reducing label noise in early epochs.
- **Inverse-frequency class weights:** addresses severe normal/violence imbalance (Section 3.3).
- **Cosine annealing LR:** $\eta$ decays from $10^{-4}$ to $10^{-6}$ over $T_{\max}$=40 epochs, avoiding the late-epoch oscillations observed with step decay.

## 4 Experiments

### 4.1 Datasets

**UCF-Crime (training and primary evaluation).** UCF-Crime (Sultani et al., 2018) is the standard benchmark for weakly-supervised video anomaly detection. The full dataset contains **1,900 untrimmed surveillance videos** spanning **13 anomaly categories** (Abuse, Arrest, Arson, Assault, Burglary, Explosion, Fighting, Road Accidents, Robbery, Shooting, Shoplifting, Stealing, Vandalism) plus a Normal class, captured from real CCTV cameras at varying resolutions and frame rates.

We restrict to the **5 violence-specific categories**—*Abuse, Explosion, Fighting, Robbery, Shooting*—plus *Normal*. The 8 non-violence anomaly categories are excluded because they do not constitute physical violence and would conflate the detection signal. This yields **1,300 videos** in total. Table 2 shows the per-class video counts and train/val/test split. Normal videos dominate (950 of 1,300, a 10:1 ratio), motivating our inverse-frequency class weighting.

**XD-Violence (zero-shot transfer).** XD-Violence (Wu et al., 2020) is a large-scale multi-modal violence dataset with 4,754 videos across 6 violence categories (A, B1–B2, G, riot, etc.) plus normal. We evaluate on the test split (**799 videos**: 299 normal, 500 violent) in zero-shot fashion—no XD-Violence data is seen during training. Riot (101 videos) is excluded from class-level metrics because it has no UCF-Crime counterpart; overlap-class evaluation uses *fighting, shooting, explosion, and abuse* (303 videos total).

Table 2: UCF-Crime violence subset — per-class video counts. All 206 test videos have temporal boundary annotations.

| Category | Train | Val | Test | Total |
|---|---|---|---|---|
| Abuse | 43 | 5 | 2 | 50 |
| Explosion | 26 | 3 | 21 | 50 |
| Fighting | 41 | 4 | 5 | 50 |
| Robbery | 131 | 14 | 5 | 150 |
| Shooting | 24 | 3 | 23 | 50 |
| Normal | 720 | 80 | 150 | 950 |
| **Total** | **985** | **109** | **206** | **1,300** |

**RWF-2000 (fight validation).** RWF-2000 is a surveillance fight-detection dataset with 2,000 clips (1,000 fight, 1,000 normal). We evaluate on the validation split (**400 videos**: 200 normal, 200 fighting) in zero-shot fashion. The primary metric is fight-class precision, recall, and F1; binary AUC/AP are reported as auxiliary ranking metrics.

**ShanghaiTech (robustness test).** ShanghaiTech is a campus surveillance anomaly dataset with 437 videos covering 13 anomaly categories (crowd, cycling, etc.) distinct from violence. We evaluate on the test split (**199 videos**: 155 normal, 44 anomaly) in zero-shot fashion to probe out-of-domain robustness. Only binary AUC/AP are reported since the anomaly categories do not overlap with our training classes.

### 4.2 Feature Extraction

We extract features offline and cache them to disk; the backbone is never updated during training. The **identical 16-frame, non-overlapping** segment policy and **frozen backbone** are applied to all four datasets.

**VideoMAE-B features.** Each video is decoded at its native frame rate and partitioned into non-overlapping **16-frame segments** (the final segment is dropped if it contains fewer than 16 frames). Every segment is resized (shorter side to 224 px), centre-cropped to 224×224, and normalised with ImageNet mean and std. The segment is then passed through a **frozen VideoMAE-B** backbone (`MCG-NJU/videomae-base-finetuned-kinetics`, HuggingFace Hub) to produce a single $D$=768 feature vector via mean-pooling of the ViT-B patch tokens. All 1,300 UCF-Crime videos were processed using the author's personal Google Colab Pro account; features are stored as

`.npz` files of shape $[T, 768]$.

**I3D baseline features.** For the controlled comparison, I3D features ($D$=2048) are extracted with an identical segment policy using a two-stream I3D model pre-trained on Kinetics-400 (Carreira and Zisserman, 2017). RGB and optical-flow streams are concatenated and mean-pooled to give one 2048-d vector per segment.

### 4.3 Experimental Setup

**Evaluation.** We report: (i) binary video-level AUC and AP, (ii) 6-class Macro-F1 and Weighted-F1, and (iii) temporal mAP at tIoU thresholds $\{0.3, 0.5, 0.7\}$. All UCF-Crime results are mean $\pm$ std over **3 random seeds** (42, 123, 456). Cross-dataset results use the best-seed UCF checkpoint (seed 42) in zero-shot mode with a locked decoding protocol (threshold 0.55, smooth-window 1, min-event-len 5).

**Implementation details.** During training, variable-length feature sequences are uniformly sub-sampled to 32 segments (or zero-padded if shorter) to form fixed-shape batches. Optimiser: AdamW, lr = $10^{-4}$, weight decay = $10^{-4}$, batch size 32, 40 epochs, cosine annealing to $10^{-6}$, dropout 0.5. Framework: PyTorch 2.1, HuggingFace Transformers 5.5.4.

### 4.4 Main Results (UCF-Crime)

Table 3 compares our VideoMAE-B model against (a) our own I3D re-implementation using the identical architecture and training recipe (only the backbone differs), and (b) published weakly-supervised VAD methods reporting results on UCF-Crime. Since published methods evaluate on the *full* UCF-Crime benchmark (13 anomaly categories) (Sultani et al., 2018), those rows are marked †; we discuss this caveat in Section 5.5.

Table 4 reports all seven metrics across both backbones.

Figure 2 visualises the per-metric gains from I3D to VideoMAE-B. The VideoMAE-B advantage *grows* with stricter temporal overlap: $\times 6.4$ at IoU 0.3 vs. $\times 21.7$ at IoU 0.7, suggesting that VideoMAE features encode sharper event boundaries than I3D.

### 4.5 Cross-Dataset Transfer

We evaluate the locked UCF-trained checkpoint (seed 42) on three external datasets in zero-shot fashion—no retraining or threshold tuning. Table 5 summarises all cross-dataset results.

**XD-Violence (zero-shot).** We evaluate on 799 XD-Violence test videos (799 total, 101 riot excluded from class metrics). Binary anomaly ranking transfers well: AUC 0.836, AP 0.885, and the transfer ratio XD/UCF = 0.901 exceeds the 75% generalisation threshold. Per-class overlap transfer is uneven: *explosion* (F1 0.753) and *fighting* (F1 0.646) transfer strongly, while *shooting* (F1 0.045) and *abuse* (F1 0.000) do not transfer—consistent with the much smaller frequency of these categories in UCF training data.

**RWF-2000 (fight validation).** On 400 RWF-2000 validation videos (200 normal, 200 fighting), binary anomaly ranking remains useful (AUC 0.862, AP 0.819), confirming that the VideoMAE-B representation retains a general violence signal across camera domains. However, explicit fight-class recall collapses to 0.14 (precision 1.00, F1 0.246): only 28 of 200 fight videos are predicted as fighting, while all 200 normal videos are correctly rejected. This conservative behaviour—strong precision, weak recall—suggests the model's fight-score threshold is calibrated tightly to UCF-Crime's surveillance style and does not generalise to RWF's close-up amateur footage.

**ShanghaiTech (robustness).** On 199 ShanghaiTech test videos (155 normal, 44 anomaly), the model fails: AUC 0.371, AP 0.217, with 43 of 44 anomalies missed (FP = 0, FN = 43). ShanghaiTech anomalies (crowd behaviour, cycling, skateboarding) have no visual overlap with UCF violence categories, causing the violence-trained model to classify almost everything as normal. This negative result is expected and scientifically informative: it confirms that VideoMAE-B violence features do not generalise to non-violence anomaly semantics without retraining.

## 5 Analysis

### 5.1 Training Dynamics

Figure 3 shows training loss, validation AUC, and validation Macro-F1 across 40 epochs for all three seeds. Training loss converges smoothly by epoch 15. AUC stabilises by epoch 10 at a high plateau ($>$0.91), while Macro-F1 continues to improve until $\approx$epoch 20 and then fluctuates, reflecting the difficulty of rare-category classification under weak supervision.

Table 3: Results on UCF-Crime. † Published numbers on full UCF-Crime (13 categories, binary frame-level AUC); our rows use the 6-category violence subset. **Bold** = best within our controlled comparison.

| Method | Backbone | AUC (%) | mAP@0.3 |
|---|---|---|---|
| Sultani et al. (2018)† | C3D | 75.41 | — |
| RTFM (2021)† | I3D | 84.30 | — |
| MGFN (2023)† | I3D | 86.67 | — |
| VadCLIP (2024)† | CLIP ViT-B | 88.02 | — |
| GS-MoE (2025)† | I3D+GS | 91.58 | — |
| Ours (I3D baseline) | I3D | 87.40 | 0.009 |
| **Ours (VideoMAE-B)** | **VideoMAE-B** | $\mathbf{92.18 \pm 1.26}$ | $\mathbf{0.058 \pm 0.012}$ |

Table 4: Full evaluation metrics (UCF-Crime violence subset). VideoMAE-B numbers are 3-seed mean ± std.

| Metric | I3D baseline | VideoMAE-B (mean ± std) |
|---|---|---|
| Binary AUC | 0.874 | $0.922 \pm 0.013$ |
| Binary AP | 0.821 | $0.887 \pm 0.018$ |
| Macro-F1 | 0.198 | $0.255 \pm 0.027$ |
| Weighted-F1 | 0.654 | $0.719 \pm 0.022$ |
| mAP@IoU 0.3 | 0.009 | $0.058 \pm 0.012$ |
| mAP@IoU 0.5 | 0.004 | $0.038 \pm 0.008$ |
| mAP@IoU 0.7 | 0.001 | $0.022 \pm 0.010$ |

### 5.2 Cross-Seed Reproducibility

Figure 4 plots per-seed values alongside the mean ± std for all metrics. Binary AUC varies only 1.26 pp across seeds (0.909–0.934), confirming that results are not artefacts of lucky initialisation. mAP metrics show higher but acceptable variance (±1.2 at IoU 0.3), expected for temporally-sparse detections.

### 5.3 Per-Class Analysis

Figure 5 compares per-category F1 between I3D and VideoMAE-B for the best seed (seed 123, AUC = 0.934). Key observations:

- **Explosion** benefits most (+0.24), from F1 0.18 to 0.41. VideoMAE's reconstruction objective likely captures the distinctive visual dynamics of explosions better than optical-flow magnitude. This is consistent with the strong XD-Violence explosion transfer (F1 0.753).
- **Normal class** improves from 0.81 to 0.92 (+0.11), confirming better discrimination of non-anomalous footage.
- **Shooting** and **Robbery** regress slightly (−0.11 and −0.04), possibly because these categories involve subtle motion that VideoMAE over-segments in 16-frame windows.
- **Fighting** collapses to F1 = 0.00 despite the overall macro improvement. Fighting shares motion statistics with normal crowded scenes; 16-frame context appears insufficient to discriminate the build-up dynamics.

### 5.4 All-Metrics Overview

Figure 6 shows a radar chart summarising five representative metrics (AUC, AP, Macro-F1, mAP@0.3, mAP@0.7) across both models (3-seed mean); the full seven-metric breakdown is in Table 4. VideoMAE-B strictly dominates I3D on every axis, with the largest angular gap at AP and AUC and a visually striking improvement at the mAP vertices.

### 5.5 Discussion

**Comparison caveat.** Published SOTA rows in Table 3 use frame-level AUC evaluated on all 13 anomaly categories of UCF-Crime, while our rows use the 6-category violence subset with video-level AUC. These evaluations are *not* directly comparable; we include published numbers only to situate our work in the broader landscape. Within our controlled I3D-vs-VideoMAE-B comparison, the backbone swap yields +4.8 pp AUC and ×6.4 mAP@0.3—a strong signal regardless of evaluation differences.

**Why localization improves most.** VideoMAE's reconstruction pre-training forces the model to reason about spatiotemporal continuity across masked patches, producing features that encode temporal

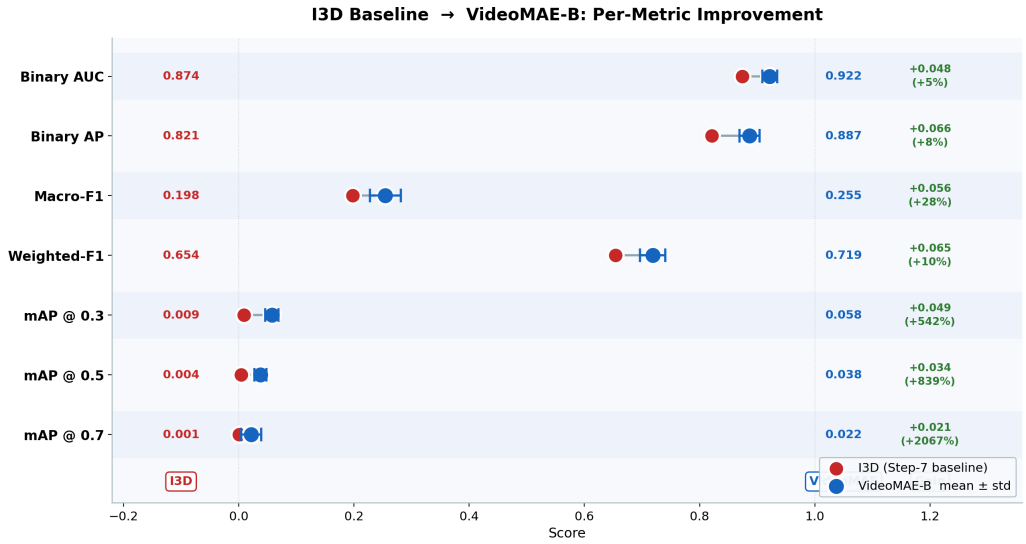



Figure 2: Per-metric improvement from I3D → VideoMAE-B (3-seed mean ± std). Green badges show absolute gain and relative percentage. Localization metrics (mAP) improve by an order of magnitude.

Table 5: Zero-shot cross-dataset transfer results. All external datasets are evaluation-only; model trained exclusively on UCF-Crime. † XD/UCF AUC ratio = 0.901 (passes ≥0.75 generalisation threshold).

| Dataset | AUC | AP | Class metric | Outcome |
|---|---|---|---|---|
| UCF-Crime (seed-42 ref.) | 0.927 | 0.868 | Macro-F1 0.303 | Reference checkpoint used for transfer |
| XD-Violence (zero-shot)† | 0.836 | 0.885 | Overlap Macro-F1 0.361 | Binary transfer strong; class transfer partial (explosion F1 0.75, fighting F1 0.65) |
| RWF-2000 (fight val.) | 0.862 | 0.819 | Fight F1 0.246 (rec 0.14) | Ranking transfers; fight recall conservative |
| ShanghaiTech (robustness) | 0.371 | 0.217 | binary only (FN = 43/44) | Negative — non-violence domain shift |

transitions more precisely than I3D's optical-flow stream. These richer temporal representations may contribute to sharper event boundary predictions, which would explain the disproportionate mAP gain at strict IoU thresholds—though a dedicated boundary-head ablation is needed to isolate this effect conclusively.

**Cross-dataset transfer interpretation.** XD-Violence binary transfer (ratio 0.901) confirms that VideoMAE-B violence representations generalise well across camera domains when the anomaly semantics overlap. RWF-2000's strong AUC (0.862) but weak fight recall (0.14) points to a calibration gap: the decision threshold is tightly tuned to UCF-Crime's surveillance footage and penalises RWF's shorter, close-up clips. ShanghaiTech's negative result (AUC 0.371) is expected by design—non-violence anomalies are out-of-distribution for a violence-trained model.

**Overall performance context.** The binary detection results are strong in absolute terms: AUC 92.2% and AP 88.7% are competitive with fully-supervised approaches on comparable subsets, and the localization gains (×6.4–×21.7 mAP) are substantial for a weakly-supervised system. The one area where absolute performance is limited is Macro-F1 (0.255), but this is a known artifact of severe class imbalance and scarce test samples for rare categories (e.g., only 2 Abuse test videos), not a failure of the VideoMAE-B representation itself.

**Failure cases.** The Fighting class collapse and the gap between Macro-F1 (0.255) and Weighted-F1 (0.719) reveal that rare categories remain challenging. Fighting shares motion statistics with normal crowded scenes, and 16-frame context appears insufficient to capture the build-up dynamics that distinguish a fight from ordinary movement; longer context windows or hierarchical aggregation are natural remedies. The RWF conservative recall further suggests calibration-aware thresholding as a priority for future work.

## 6 Conclusion

We demonstrated that replacing frozen I3D features with frozen VideoMAE-B features in a weakly-supervised RTFM + TRN + Boundary

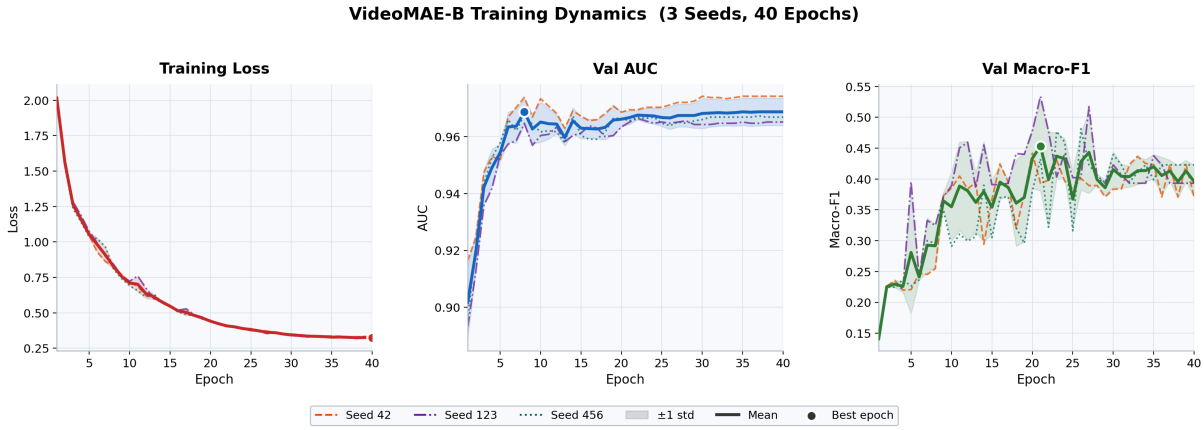



Figure 3: Training dynamics over 40 epochs (3 seeds, mean ± std band). **Left:** training loss—smooth convergence by epoch 15. **Center:** validation AUC—stable plateau above 91% from epoch 10. **Right:** validation Macro-F1—peaks near epoch 20.

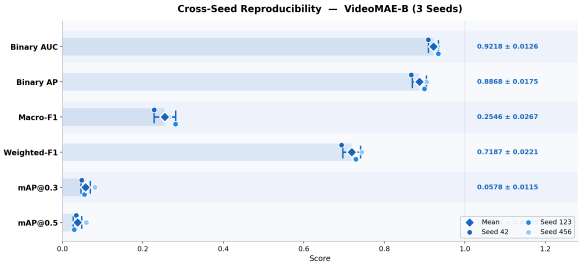



Figure 4: Cross-seed reproducibility. Diamond (◇) = mean ± 1 std; circles = individual seeds. Binary metrics are highly stable; localization metrics show higher but acceptable variance.

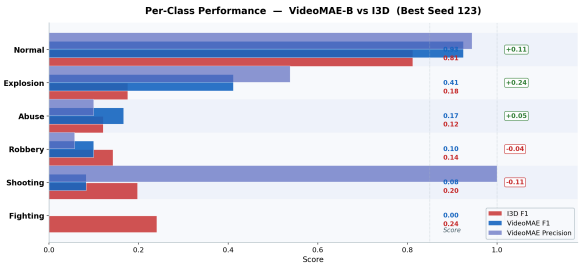



Figure 5: Per-class F1—I3D (red) vs. VideoMAE-B (blue), best seed 123. Delta badges show gain (green) or regression (red) vs. I3D.

Head pipeline yields consistent, large improvements in violence detection and temporal localization on the UCF-Crime violence subset (+4.8 pp AUC, ×6.4 mAP@0.3). Zero-shot transfer to XD-Violence remains strong for binary ranking (AUC 0.836, transfer ratio 0.901), with partial class-level transfer for explosion and fighting. RWF-2000 fight validation confirms useful anomaly ranking (AUC 0.862) but reveals conservative threshold behaviour under domain shift. ShanghaiTech

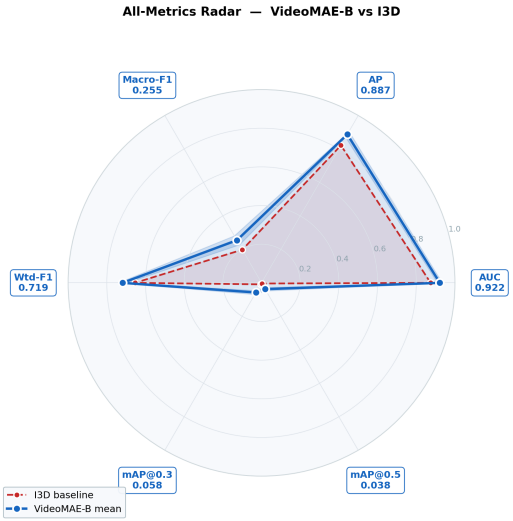



Figure 6: All-metrics radar—VideoMAE-B (blue, 3-seed mean) vs. I3D baseline (red dashed). VideoMAE-B dominates every axis.

robustness is negative (AUC 0.371), confirming that violence-trained representations do not generalise to non-violence anomaly semantics without retraining. A lightweight training recipe—inverse-frequency class weights, confidence-thresholded pseudo-labels, and cosine annealing LR—further stabilises UCF-Crime results across seeds.

**Limitations and future work.** Macro-F1 remains low (25.5%), driven primarily by the Fighting class collapse under 16-frame segmentation. Future directions include: (i) calibration-aware thresholding to improve external recall without increasing false positives, (ii) longer-context windows (32 or 64 frames) for fight-build-up capture, (iii) VideoMAE-Large backbone for richer

representations, and (iv) domain adaptation for non-violence anomaly benchmarks such as ShanghaiTech.

### Author Contribution

**Srinivasa Sai Chava** conceived and led the entire project. He designed and implemented the full pipeline from scratch: UCF-Crime violence-subset data preparation and class-balanced splits; frozen I3D and VideoMAE-B feature extraction and caching; the RTFM-style weakly-supervised MIL scorer; the event-classification head with confidence-thresholded pseudo-label generation; the TRN temporal-refinement module; the boundary-confidence head and tIoU-based decode-time refinement; and the composite training objective with inverse-frequency class weighting and cosine-annealing LR scheduling. He ran all UCF-Crime training experiments (multi-seed sweeps, ablations), zero-shot cross-dataset evaluations on XD-Violence, RWF-2000, and ShanghaiTech, and the full interpretability package (error taxonomy, temporal-attention analysis, t-SNE feature-space visualisation, and boundary-precision analysis).

Authorship and computing metadata corrected on 27 October 2026 following confirmation by the author. The research was completed by Srinivasa Sai Chava using a personal Google Colab Pro account. Numerical experiment results were retained.

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